

Numbers ke Prakar

Types of Numbers - Aasan Hinglish Notes, Examples aur Practice Q's

Maths me numbers ko alag-alag families me baanta jata hai, aur har family pichli wali ke upar bani hoti hai. Is guide me har type ko simple Hinglish me examples ke saath samjhaya gaya hai, aur end me 22 practice questions (answer key ke saath) diye gaye hain.

1. Natural Numbers (N)

Definition: Ginti (counting) wale numbers jo 1 se shuru hote hain aur aage badhte rehte hain: 1, 2, 3, 4, ...

Symbol: N

Example: 1, 2, 3, 4, 5, ... (jaise class me students ki sankhya)

Yaad rakho:

- Sabse chhota natural number 1 hai.
- Zero (0) ek natural number NAHI hai.
- Ye hamesha positive whole numbers hote hain.

2. Whole Numbers (W)

Definition: Saare natural numbers + 0 milakar: 0, 1, 2, 3, 4, ...

Symbol: W

Example: 0, 1, 2, 3, 4, ...

Yaad rakho:

- Whole numbers = Natural numbers + 0.
- Sabse chhota whole number 0 hai.
- Har natural number ek whole number hai (par 0 whole hai, natural nahi).

3. Integers (Z)

Definition: Saare whole numbers + unke negatives: ..., -3, -2, -1, 0, 1, 2, 3, ...

Symbol: Z

Example: -100, -5, -1, 0, 3, 47

Yaad rakho:

- Isme positive, negative aur zero teeno aate hain.
- Integers me koi fraction ya decimal nahi hota.
- Zero ek integer hai jo na positive hai na negative.

4. Rational Numbers (Q)

Definition: Wo numbers jinhe p/q fraction ke roop me likha ja sake, jahan p aur q integers hon aur q zero ke barabar na ho.

Symbol: Q

Example: $1/2$, $-3/4$, 5 ($= 5/1$), 0.75, 0.333... ($= 1/3$)

Yaad rakho:

- Jo decimals khatam ho jayein (terminating) ya repeat hon, wo rational hain.
- Har integer bhi ek rational number hai.
- Denominator q kabhi 0 nahi ho sakta.

5. Irrational Numbers

Definition Wo numbers jinhe simple fraction p/q me NAHI likha ja sakta. Inka decimal part kabhi khatam nahi hota aur repeat bhi nahi karta.

Example: $\pi = 3.14159\dots$, $\sqrt{2} = 1.41421\dots$, $e = 2.71828\dots$

Yaad rakho:

- Ye non-terminating aur non-repeating decimals hote hain.
- Non-perfect squares ke square root irrational hote hain ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$).
- π aur e sabse famous irrational numbers hain.

6. Real Numbers (R)

Definition Saare rational aur irrational numbers milakar. Jo bhi number number-line par rakha ja sake, wo real number hai.

Symbol: $\mathbb{R}$

Example: -2, 0, $1/2$, 7.5, $\sqrt{2}$, π

Yaad rakho:

- Real numbers = Rational numbers + Irrational numbers.
- Roz-marra ke lagbhag saare numbers real numbers hote hain.
- $\sqrt{-1}$ jaise numbers real NAHI hote - unhe imaginary numbers kehte hain.

7. Even aur Odd Numbers

Definition Even numbers 2 se पूरी तरह divide ho jate hain; odd numbers 2 se divide nahi hote.

Example: Even: 2, 4, 6, 8, 10 | Odd: 1, 3, 5, 7, 9

Yaad rakho:

- Even numbers ke end me 0, 2, 4, 6 ya 8 aata hai.
- Odd numbers ke end me 1, 3, 5, 7 ya 9 aata hai.
- Zero (0) ek even number hai.

8. Prime aur Composite Numbers

Definition Prime number ke exactly do factors hote hain (1 aur khud wo number). Composite number ke do se zyada factors hote hain.

Example: Prime: 2, 3, 5, 7, 11 | Composite: 4, 6, 8, 9, 10

Yaad rakho:

- 2 hi ekmatra even prime number hai.
- 1 na prime hai na composite.
- Sabse chhota prime 2 hai; sabse chhota composite 4 hai.

Quick Revision Summary

Number families ek dusre ke andar aise fit hoti hain:

Natural < Whole < Integers < Rational < Real

Irrational numbers bhi Real hain, par Rational se alag.

Real Numbers = Rational + Irrational (number line ke saare numbers).

Practice Questions (Khud Try Karo!)

In 22 questions ko khud solve karne ki koshish karo. Answers neeche answer key me diye gaye hain.

- Q1. Sabse chhota natural number kaunsa hai?
- Q2. Sabse chhota whole number kaunsa hai?
- Q3. Kya 0 ek natural number hai? (Haan/Nahi)
- Q4. Kya 0 ek whole number hai? (Haan/Nahi)
- Q5. -5 kis type ka number hai (natural / whole / integer)?
- Q6. Kya har natural number ek whole number hota hai?
- Q7. $\frac{3}{4}$ kis type ka number hai?
- Q8. $\sqrt{2}$ rational hai ya irrational?
- Q9. π (3.14159...) rational hai ya irrational?
- Q10. 7 ko p/q form me likho.
- Q11. Kya 0.75 rational hai? (Haan/Nahi)
- Q12. Sabse chhota prime number kaunsa hai?
- Q13. Kya 1 prime hai ya composite?
- Q14. 2 ke alawa koi aur even prime number hai? (Haan/Nahi)
- Q15. 4, 6 aur 8 - ye prime hain ya composite?
- Q16. Kya 0 even hai ya odd?
- Q17. 17 prime hai ya composite?
- Q18. Real numbers kis-kis se milkar bante hain?
- Q19. Kya $\sqrt{-1}$ ek real number hai?
- Q20. -3, 0 aur 5 - ye sab kis ek hi category me aate hain?
- Q21. 9 ke saare factors likho. Kya 9 composite hai?
- Q22. 0.333... (repeating) rational hai ya irrational?

Answer Key

- Q1: 1 (sabse chhota natural number)
- Q2: 0 (sabse chhota whole number)
- Q3: Nahi (0 natural number nahi hai)
- Q4: Haan (0 whole number hai)
- Q5: Integer (negative hone ke kaaran natural/whole nahi)
- Q6: Haan (par 0 whole hai, natural nahi)
- Q7: Rational number (p/q form me likha ja sakta hai)
- Q8: Irrational (p/q form me nahi likh sakte)
- Q9: Irrational (non-terminating, non-repeating)
- Q10: $\frac{7}{1}$
- Q11: Haan (0.75 terminating decimal hai)
- Q12: 2
- Q13: Dono nahi - 1 na prime hai na composite
- Q14: Nahi - 2 hi ekmatra even prime hai
- Q15: Composite (sabke 2 se zyada factors hain)

- Q16: Even (0 even number hai)
- Q17: Prime (factors sirf 1 aur 17)
- Q18: Rational + Irrational numbers
- Q19: Nahi (ye imaginary number hai)
- Q20: Integers
- Q21: Factors: 1, 3, 9 - Haan, 9 composite hai (2 se zyada factors)
- Q22: Rational (repeating decimal = $\frac{1}{3}$)